EXP NO:17 EYE DIAGRAM ANALYSIS OF OPTICAL COMMUNICATION LINK USING OPTISYSTEM

OBJECTIVES

1. Design and establish a fiber optic communication link using OptiSystem software.

2. Measure and evaluate the Bit Error Rate (BER) of the system for different fiber lengths.

3. Study the effect of fiber length on BER, Q-factor, a enuation, and dispersion.

EXPECTED OUTCOMES

1. Designed and simulated a fiber optic communication link in OptiSystem.

2. Observed and analyzed transmi ed and received optical signals across different transmission distances.

3. Measured BER and Q-factor for varying fiber lengths.

4. Compared performance across short, medium, long, and very long fiber links

SOFTWARE REQUIRED

Optisystem software

OBJECTIVE LINKAGE:

The experiment provides practical understanding of optical fiber link design and performance analysis using OptiSystem by evaluating BER for different fiber lengths and studying their effects on Q-factor, attenuation, and dispersion. It reinforces concepts of link diagnostics, signal degradation, dispersion, BER performance, and optical link design.

EXPERIMENT PROCEDURE :

1. Open OptiSystem and create a new project.

2. Add a PRBS Generator as the data source.

3. Connect the PRBS Generator to an NRZ Pulse Generator.

4. Add a CW Laser Source for optical transmission.

5. Connect the laser output to the optical fiber.

6. Set the transmi er parameters such as wavelength, bit rate, and optical power.

7. Set the optical fiber length according to the required test case.

8. Keep a enuation and dispersion values as specified in the fiber parameters.

9. Connect a PIN Photodetector at the receiver.

10. Add a Low Pass Filter and 3R Regenerator after the photodetector.

11. Connect a BER Analyzer to the receiver output.

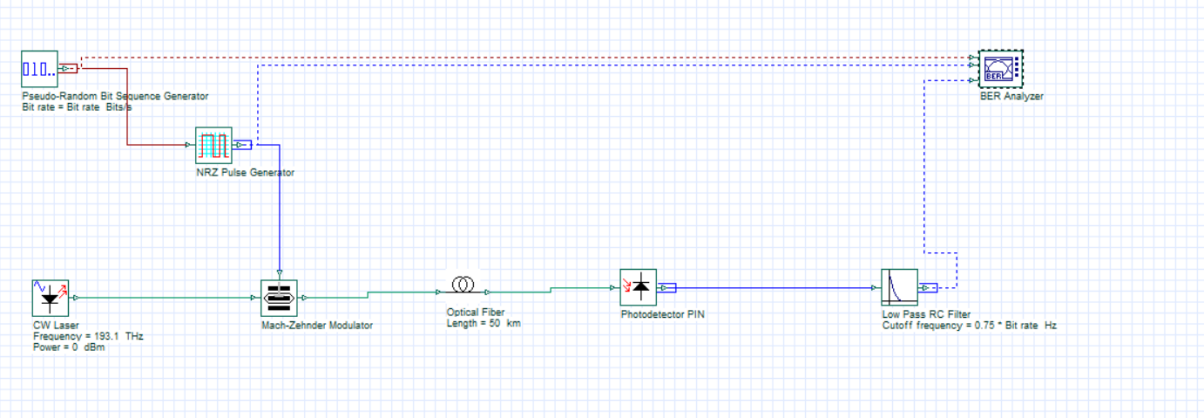
12. Run the simulation for each fiber length.

13. Observe the BER, Q-factor, and Eye Diagram.

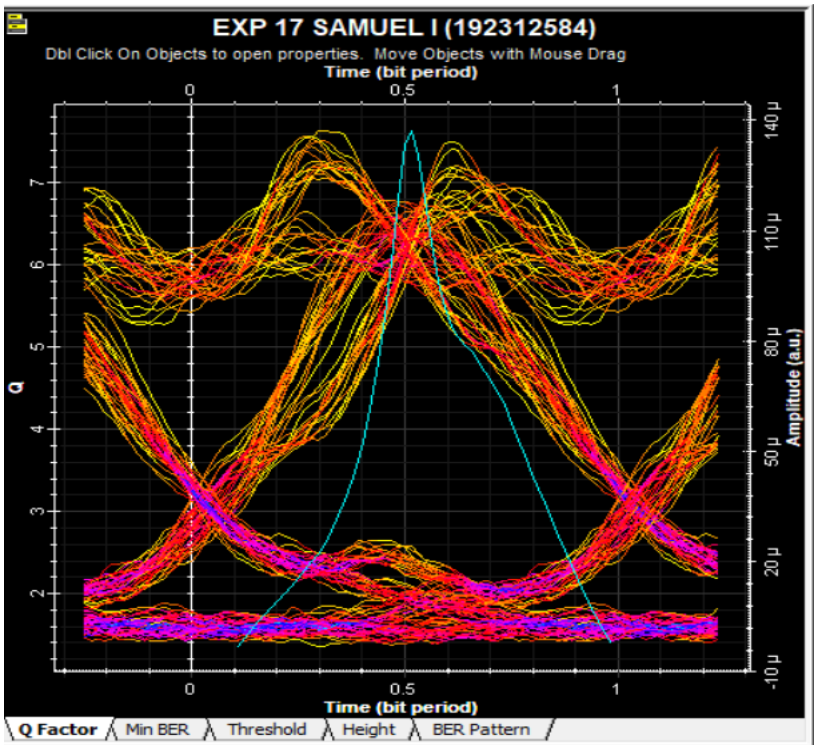
14. Compare the performance for different fiber lengths.

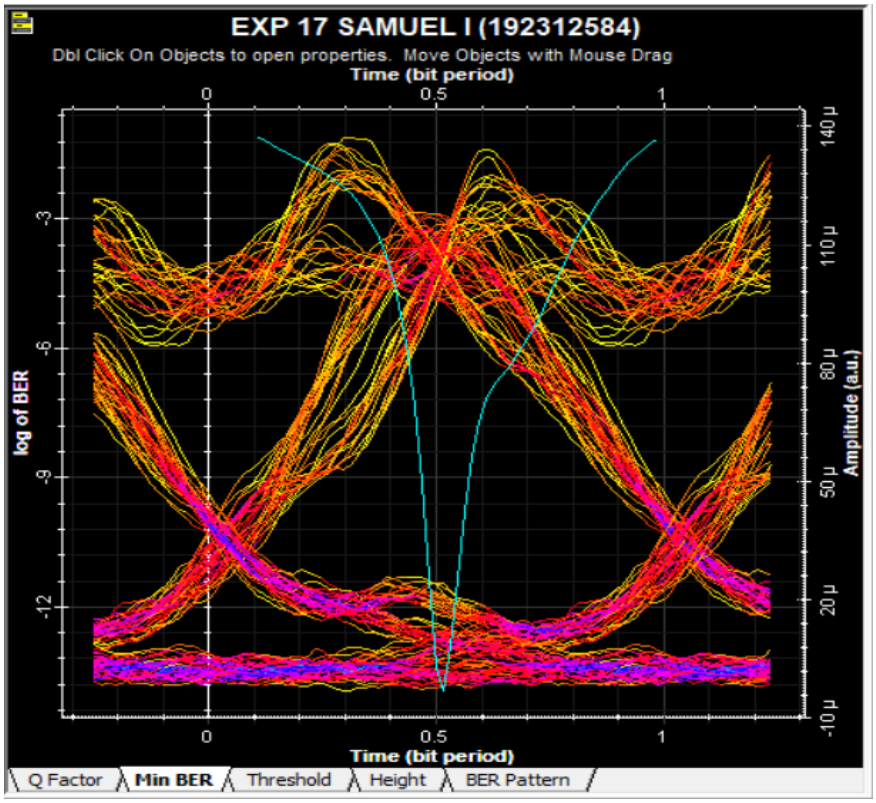
15. Save the results and conclude the experiment.

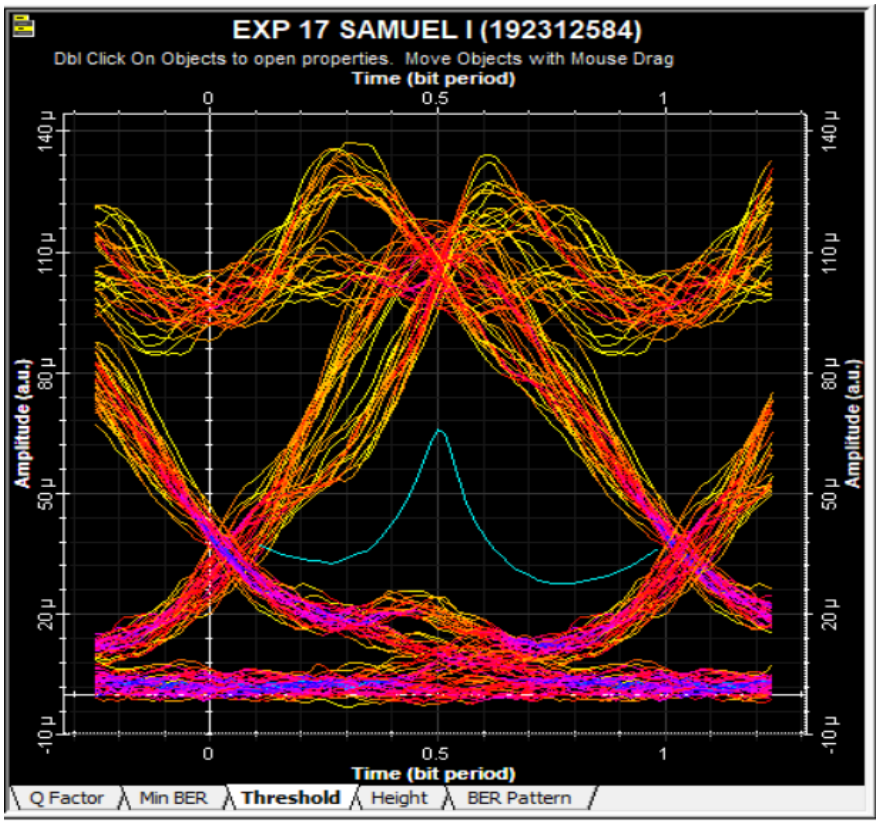
BLOCK DIAGRAM

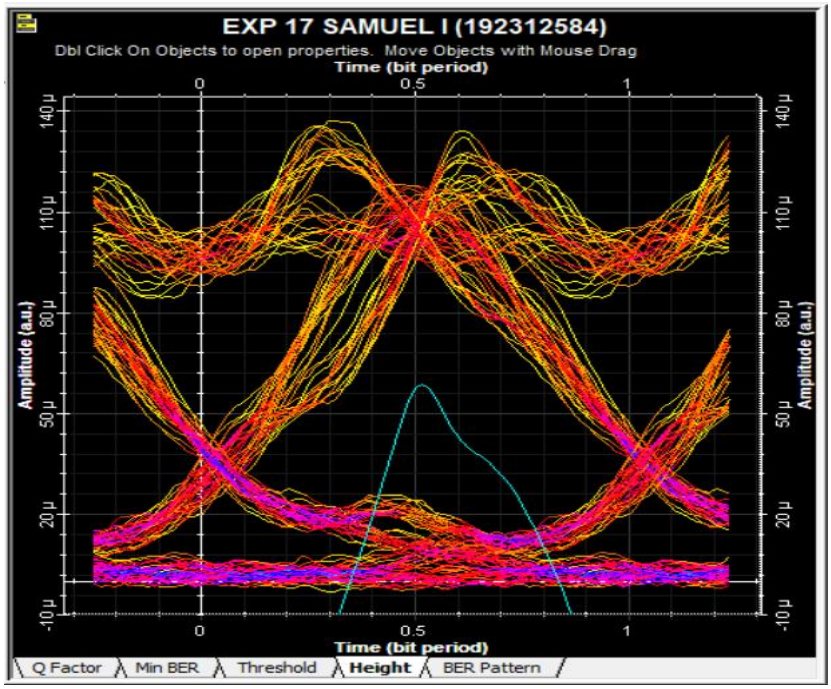


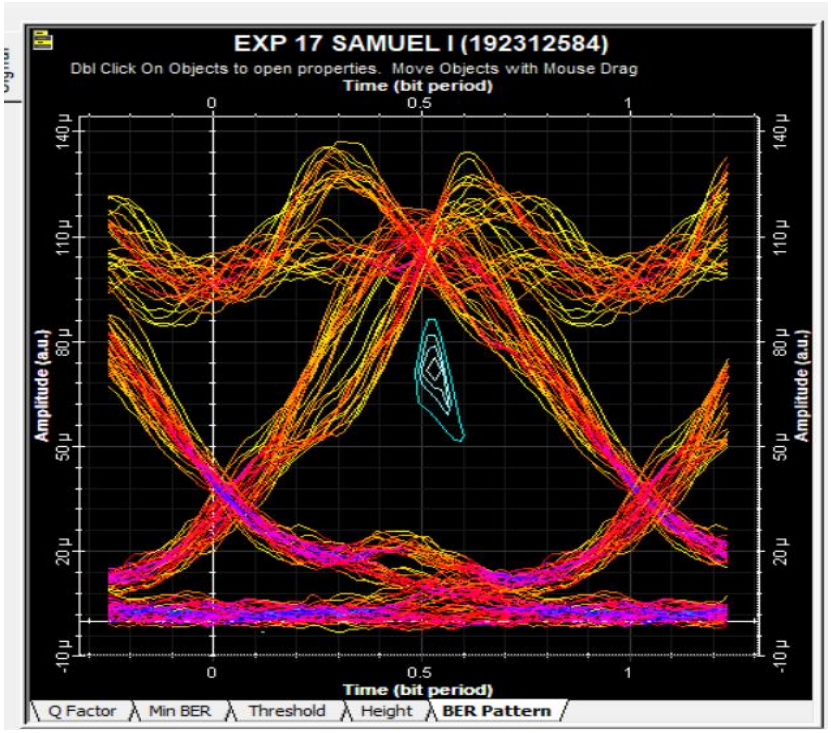
OUTPUT :




EXP 17 SAMUEL I (192312584)
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
0
0.5
1
-3
-6
-9
-12
log of BER
140 µ
110 µ
80 µ
50 µ
20 µ
-10 µ
Amplitude (a.u.)
Time (bit period)
Q Factor
Min BER
Threshold
Height
BER Pattern


EXP 17 SAMUEL I (192312584)
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
0
0.5
1
140 µ
110 µ
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Amplitude (a.u.)
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CONCLUSION

The study shows how fiber length affects BER and signal quality in an optical communication system.